

Bias by Default

Neocolonial Visual Vocabularies in AI Image Generating Design Practices

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ABSTRACT

The growing popularity of AI image generators solidifies a novel form of co-creation and image-making while concurrently unveiling concerns regarding the representational harms engendered by these models. By examining images generated by the prompt “imagine a religious person” it becomes evident how certain aspects of the predominantly Christian patriarchal culture of the West overwhelmingly dominate the generated images. Expanding the inquiry by incorporating diverse countries into the same prompt, the comparative outcomes of the images highlight an amplification of biases combined with cultural hybridities. As a critical exercise, this research employs the concept of data colonialism to prompt reflection on how social hierarchies and linguistic ideologies shape the default visual vocabulary of these models. Consequently, the article suggests that a decolonial approach to design can broaden discussions about biases generated by these generative models in dynamic social contexts, while simultaneously offering novel insights for design practice.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models.

KEYWORDS

Visual Design, Data Colonialism, Bias, AI Image Generation

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1 INTRODUCTION

The painting “First Mass in Brazil” by Brazilian artist Victor Meirelles (Paris, 1859-1861) depicts the first Christian Catholic ceremony held at the land that would later be known as Brazil. The inspiration for this work is drawn from the letter written by Pero

Vaz de Caminha to King D Manuel (1469-1521)¹. This letter is considered the first official record of the accounts by the scribe of Pedro Álvares Cabral’s fleet, marking the discovery of new lands and the colonization process of Brazil in 1500. The lighting in the painting highlights central elements: the cross, the altar, and the newly arrived Portuguese, while on the periphery – and in the shadows – are the indigenous peoples observing the religious ritual.

Ironically, five centuries later, the examination of images generated by AI from a textual description (prompt): “/imagine: a religious person in Brazil” tends to default to the same elements that confirm the Eurocentric, patriarchal, and Christian imagery of what has become a popular tool for imaginative co-creation between humans and AI.

In that sense, our research begins with the following question: *Is it possible that generative text-to-image AI models perpetuate the same historical Eurocentric narratives, thereby generating a neocolonial visual vocabulary?*

The growing popularity and accessibility of AI image generators consolidate ‘image-making’ in co-creation with AI as another modus operandi for artists and designers. This situation simultaneously introduces a new visual language in creative design practices, raising various questions in a complex dance involving power relations, social hierarchies, linguistic ideologies, and creative autonomy. Given that AI image generators are not academic projects but rather expansive businesses in various sectors worldwide, it is opportune to investigate how the relationship between humans and AI essentially reveals itself as a power dynamic [1]. By employing the concept of Data Colonialism [2], we can analyze how technology, particularly concerning data, and consequently, AI, operates within the continuity of a colonial matrix of power, guiding stakeholders and developments.

Delving into the intricacies of AI-generated imagery, this article draws upon insights obtained from an experimental exploration of Midjourney. Divided into two phases, the investigation initially employs images intentionally generated by the generic prompt: “imagine: a religious person” to analyze potential biases related to gender and age in the context of religion. Additionally, it aims to identify cultural aspects informed by consistent visual patterns. Following the same logic, in a subsequent phase, four diverse nationalities are introduced to the prompt, maintaining the same analytical dimensions as the first phase. Comparative results underscore the expressive tendency of the generative model to portray a religious person as ‘an old male adult with a white beard and white hair, adorned with Christian symbolic elements’; these findings

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¹Carta de Pêro Vaz de Caminha. (Letter from Pêro Vaz de Caminha). Retrieved September 29, 2023 from <https://digitarq.arquivos.pt/details?id=4185836>

persist when different nationalities are added, further amplifying gender biases.

As a critical exercise, the investigation exploring the bias-by-default effect from Midjourney serves to enrich dialogues surrounding the nuanced relationships existing between AI, data, and colonialism. Simultaneously, it sheds light on the persistent entrenchment of design practices in Western epistemologies, a phenomenon extending its influence globally, even within developing nations, often neglectful of local knowledge systems [3]. This intersection, particularly with the thematic backdrop of religion, not only underscores the intrinsic ontological challenges embedded in AI [1] [4] [5] but also unfurls new avenues for discourse on power dynamics, inclusivity, and the imperative need for decolonization in both research and technology design. These discussions are paramount in addressing contemporary global challenges, such as racial justice, economic inequality, and political dynamics, thereby resonating with the concerns and aspirations of the broader Human-Computer Interaction community [6] [7] [8].

2 BACKGROUND

2.1 Data Colonialism

In May 2017, the cover of *The Economist* portrayed data as the most valuable resource in the world, metaphorically comparing it to oil². However, it is well-known that data does not present itself in nature as readily available raw material. Data is intricately tied to human relationships in society, and as such, it needs to be appropriated [2].

Consequently, the gradual transformation of social relations into data relations occurs within profit-oriented practices, predominantly controlled by major private technology corporations. This new social order can be better understood through the lens of historical colonialism, where economic power relations (that generate value) and cognitive power relations (that concentrate knowledge) converge. Thus, data colonialism is not merely a form of resource appropriation but also a means of acquiring the resources to understand and organize the world from specific centers of power. This condition perpetuates the same injustices affecting bodies and communities globally, reinforcing marginalization and inequalities between the "Global North" and "Global South" [9] [10] [11].

Similar to historical colonialism, which appropriated territories and resources to govern subjects, data colonialism, embedded in capitalist practices, unilaterally claims human experience as free raw material for translation into data. This process fuels advanced manufacturing processes known as 'machine intelligence,' resulting in the production of prediction products [12]. This condition, reflected in the context of AI models, presents unprecedented opportunities for social discrimination, behavioral influence, and economic inequality. Consequently, it fosters resistance to emerging power dynamics, addressing issues such as algorithmic bias [13] [14] [15], the erosion of human autonomy [16], and unjust labor exploitation [17] [18] [19] while reinforcing disparities in returns obtained by Global South [19] [20].

²Retrieved September 29, 2023 from <https://www.economist.com/leaders/2017/05/06/the-worlds-most-valuable-resource-is-no-longer-oil-but-data>

2.2 Generative AI and Coloniality

2.2.1 Text-to-image generative models. As conventionally defined, artificial intelligence (AI) focuses on the development of computers capable of engaging in human-like thought processes. These processes include learning, reasoning, and self-correction [21]. Broadly, there are two types of AI: deductive and generative.

Deductive AI involves analyzing large datasets to identify patterns, while generative AI produces new content by learning from existing data [20]. Recent advances in generative AI, particularly in Text-to-Image models (TTI), have gained popularity for crafting visual content from Natural Language Processes (NLP) without users needing to understand the underlying technology [22]. These models leverage neural networks trained on extensive datasets of images and descriptions, with diffusion models being the current leading techniques, achieving photorealistic quality and generalizing to unseen images [23].

The rapid improvement in the visual outputs produced by TTI models has expanded their applications in arts, design, and communication. Models like Midjourney and DALL-E are gaining popularity with an increasing number of users and communities globally³. While this creates conditions for new opportunities and developments, concerns about risks and potential harms generated by these models are also on the rise.

2.2.2 NLP, Bias and Power Dynamics. A critical survey of 146 papers on bias in NLP conducted by Blodgett et al. underscores the importance of questioning how existing social hierarchies and linguistic ideologies are driving forces in the development and deployment of NLP systems, consequently AI image generators [24]. It further explores how these systems, in turn, reproduce these same hierarchies and ideologies, reinforcing bias as a default feature of these models.

Building on this premise, Bird et al. suggest that the analysis of the stakeholders involved in the processes can unravel the power dynamics reproducing risks and harms of discrimination, exclusion, misuse, misinformation, and disinformation [25]. It includes the examinations of the relationships among System Developers, Data Sources, Data Subjects, Users, Affected parties, and Regulators.

2.2.3 Language e Coloniality. Large Language Models (LLM) technologies, when applied in AI image generator models, interpret user prompts, translating the semantic meaning into a numerical vector that guides the diffusion process. LLMs are not models of brains but of language itself, encompassing its patterns, structures, and probabilities [26]. If we consider that language is the means by which social groups are labeled and a way in which beliefs about social groups are transmitted, group labels can serve as the basis for the formation of stereotypes and, thus, reinforce social inequalities [24].

Therefore, the content of the label functions to identify a specific category of people, conveying category boundaries and a position

³Since its establishment in July 2022, the Midjourney platform has garnered over 17 million users, with Brazil ranking as the second-largest country in terms of user numbers. Discord and Photutorial. Retrieved December 10, 2023 from <https://discord.com/servers/midjourney-662267976984297473> and <https://photutorial.com/midjourney-statistics>

OpenAI's DALL-E has over 1.5 million users, generating more than 2 million images per day. Open-AI. Retrieved December 10, 2023 from <https://openai.com/blog/dall-e-now-available-without-waitlist>



Figure 1: First mass in Brazil, Victor Meirelles. 1859-61, Paris. Oil painting on canvas.

in a hierarchical taxonomy. In this context, when such generative AI models reinforce stereotypes [27] [28] or simply fail to recognize certain social groups [29] [30], this can be perceived as an exercise of a coloniality of power [31] [32].

A colonial power matrix serves as an organizing principle that involves the exercise of exploitation and domination in multiple dimensions of social life, from the economic, sexual, or gender relations to political organizations, knowledge structures, state institutions, and family units [31]. In the context of AI, this position challenges us to reflect on how generative models broadly incorporate the racial, gender, spiritual, linguistic, economic, political, and epistemic hierarchies of a modern eurocentric colonial world-system.

The continuum of a colonial power matrix that interconnects the past, present, and potential future can be evidenced in the visual representations of the colonizer's acts, such as the painting of the first mass held in Brazil by the Portuguese in 1500 (Figure 1), and also in the outputs of the Midjourney algorithms in 2023, where a religious mark is affixed to the forehead of an individual seated on the ground in a street (Figure 2).

2.3 Decolonial design practices

The AI black-box condition has often required Human-Computer Interaction (HCI) expertise to incorporate the human aspect into the evaluation of AI systems regarding bias in their outputs [33] [34]. It has also been observed that HCI practices inherently promote neocolonial design, raising concerns within HCI about a lack of cross-cultural sensitivity and the problematization of power relations with users from previously colonized contexts [3].

On the other hand, a decolonial perspective points out that design practices are far from universal and still rooted in Western epistemologies, being applied in countries like developing nations without sufficient consideration for local knowledge systems [3]. Therefore, the discourse on decolonizing design is ontological in nature and requires a transcultural and transdisciplinary approach to radically transform design [35].



Figure 2: A religious person in Brazil. 2023. Generated by Midjourney.

In the investigation involving Midjourney [30], Allahyari instructed the model to imagine “a woman with a hijab next to a woman without a hijab on the streets of Tehran”. Despite centuries of struggles by Iranian women for the right to choose or have agency regarding the hijab, the model failed to generate any images of women without hijabs. This situation reveals how language and the semantics of culture, religion, feminism, and history still pose challenges for generative AI models, establishing power dynamics that limit the artist/design-user in designing possible or alternative futures.

While Participatory Design (PD) approaches traditionally address issues related to power relations, a gap is evident concerning cultural power relations [36]. The full potential of PD lies in integration with other approaches, with a deeper attention and understanding of cultural realities and the ontological entanglements of everyday practices when working with non-Western people, as well as minority communities and stakeholders [3]. Therefore, emerging HCI methods, techniques, and approaches need to be attuned to the sense of ‘in-betweenness’ necessary for the development of appropriate computing technologies in cultural spaces [36].

3 METHODOLOGY

3.1 AI model data generation and generic prompts

In active research, the Midjourney platform, version 5.2, was utilized, employing an image-based Generative Adversarial Neural Network (GAN). Midjourney integrates large language models (LLM) and diffusion models. The LLM interprets user-input prompts semantically, converting them into numerical vectors. These vectors guide the diffusion process, gradually introducing random noise to a computer's image training dataset.



Figure 3: Phase 1 – Control Group. Prompt: “/image a religious person”



Figure 4: Phase 2 – Nationalities Group. Prompt: “/image a religious person in X”. X= Brazil, USA, Nigeria, Japan

Through iterative learning, the model learns to reverse this noise, recovering the original image and gaining the capability to generate new images. The architecture and training data of Midjourney remain undisclosed. Operated through Discord chat, Midjourney defaults to generating four images for each user prompt. The investigation collected images following this procedure.

For the generation of images, generic prompts containing only a few words were employed within the structure: “Imagine a religious person” and “Imagine a religious person in X,” where X represents a country. As previously noted, the utilization of generic prompts [27] is justified by the intention to assess the model’s ability to abstract social and/or diverse concepts like “person”, “religion”, “country”, revealing (or not) potentially biased results.

3.2 Strategy and structure

The investigation unfolds in two phases. In the Phase 1 – Control Group, the collection of the initial 100 images is conducted based on the prompt “imagine a religious person” (Figure 3). The outputs are then analyzed in 2 dimensions: perceived gender and age. The analysis dimensions are applied based on general criteria, aiming to detect only expressive similarities informed by visual elements present in the generated images.

In the Phase 2 – Nationalities Group, 400 images are then collected and evenly distributed among four distinct groups using the prompt “imagine a religious person from X,” where X represents Brazil, the USA, Nigeria, and Japan (Figure 4). The selection of countries is based on religious diversity, population size, and geographical position. For each of the four groups, the same analysis

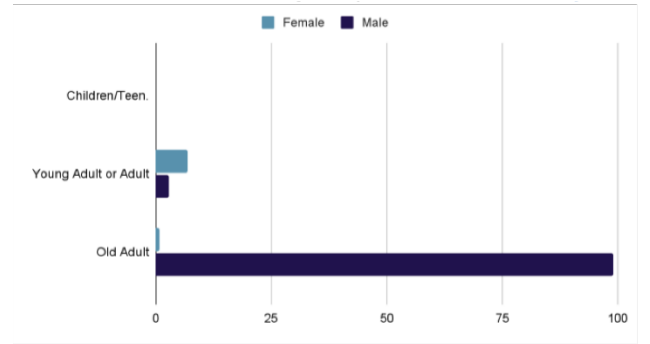


Figure 5: Control Group. Age and Gender Analysis Chart.

dimensions as the control group are applied. Subsequently, a comparative analysis is conducted between the four groups and the control group.

3.3 Results

3.3.1 Phase 1 – Control Group.

- **Gender⁴** – 92% of the images depict Male figures, and only 8% Female figures.
- **Age⁵** – 90% of images depict Old Adults (98,8% Male / 1,1% Female). 10% depict Adults/Yr. (70% Female / 30% Male). No images depict Children/Teen (Figure 5).

⁴In order to classify gender bias within representational harm, we consider Male and Female as genders perceived in a generic way.

⁵We divide ages into 3 generic classification groups: 1. Children/Teen (0-18 years old); 2. Adult/Yr (19-64 years old); 3. Old Adult (+65 years old).



Figure 6: A religious person. 2023. Generated by Midjourney.

Other observations:

- *Facial expressions* — 99% of the images display a person with serious expressions, and only 1% exhibit a slight smile (Adult/Yr. Female).
- *Physical features* — 100% of Old Adult Male exhibit White Hair and White Beard (Figure 6).
- *Prominent symbols* — 24% of the images feature the Christian Cross in clothing, or as an artifact.

3.3.2 Phase 2 – Nationalities Group. **Brazil:**

- *Gender* — 100% of the images depict Male figures. (Females represent 50,9% of Brazil's population)⁶
- *Age* — 74% of images depict Old Adults (Male). 26% of Adults/Yr. (Male). No images depict Children/Teen.

Other observations:

- *Physical features* — 83% of Old Adult Males exhibit White Hair and White Beard. 10% depict Male with facial tattoos.
- *Prominent symbols* — 22% of the images contain the Christian Cross, in 7% it is displayed on the face, in the form of a tattoo, painting or attached element.

USA:

- *Gender* — 93% of the images depict Male figures, and 7% depict Females figures.
- *Age* — 92% of images depict Old Adults (99% Male / 1% Female). 8% of Adults/Yr. (87,5% Female / 12,5% Male). No images depict Children/Teen.

Other observations:

⁶Total population of Brazil from 2008 to 2022, by gender (in million inhabitants). Statista. Retrieved December 10, 2023 from <https://www.statista.com/statistics/788353/population-total-gender-brazil/#:~:text=In%202022%2C%20the%20total%20population,in%20this%20Latin%20American%20country.>

- *Physical features* — 100% of Old Adult Males exhibit White Hair and White Beard.
- *Prominent symbols* — 94% of the images contain the USA National flag. One image features a woman dressed as the Statue of Liberty. In another image an Old Male Adult wears a similar Ku Klux Klan outfit.

Nigeria:

- *Gender:* 100% of the images depict Male figures. (Females represent 49% of Nigeria's population)⁷.
- *Age:* 82% of images depict Old Adults (Male). 18% of Adults/Yr. (Male). No images depict Children/Teen.

Other observations:

- *Physical features* — 86% of Old Adult Males exhibit White Hair and White Beard.
- *Prominent symbols* — Even in a country where the majority of the population is declared Muslim⁸, the majority of images do not feature any Islamic symbol. The Christian Cross appears in 22% of images and elements of Christianity also stand out in religious clothing and artifacts.

Japan:

- *Gender:* 99% of the images depict Male figures, and 1% depict Female figures.
- *Age:* 95% of images depict Old Adults (100% Male). 5% of Adults/Yr. (80% Male / 20% Female). No images depict Children/Teen.

Other observations:

- *Physical features* — 94% of Old Adult Males exhibit White Hair and White Beard.
- *Prominent symbols* — The images do not portray any symbol of the Shinto religion, considered the largest religion in Japan⁹. Even though Christianity represents only 1.5% of the population, 7% of images feature the Christian Cross in the image.

The comparative analysis among the four nationalities (Brazil, USA, Nigeria, and Japan) reveals significant age differences, with a predominant representation of older adults, as depicted in Figure 7. Additionally, a marked gender disparity is observed, predominantly favoring males, as illustrated in Figure 8.

4 FINDINGS AND DISCUSSION

The comparative results among the samples of over 500 images collected in this critical exercise initially show that for the generative model of Midjourney, a “religious person” is typically an old adult male, with a white beard and white hair, surrounded by European Christian-like elements (Figure 9).

When adding nationality to the prompt, gender biases are amplified, sometimes with the complete absence of female figures (Brazil and Nigeria). For the four additional nationalities, the same

⁷Total population in Nigeria from 2011 to 2021, by gender (in millions). Statista. Retrieved October 20, 2023 from

<https://www.statista.com/statistics/967908/total-population-of-nigeria-by-gender/>

⁸Harvard Divinity School. Religion and Public Life. Retrieved October 11, 2023 from <https://rpl.hds.harvard.edu/faq/islam-nigeria>

⁹Wikipedia. Religion in Japan. Retrieved October 06, 2023 from https://en.wikipedia.org/wiki/Religion_in_Japan

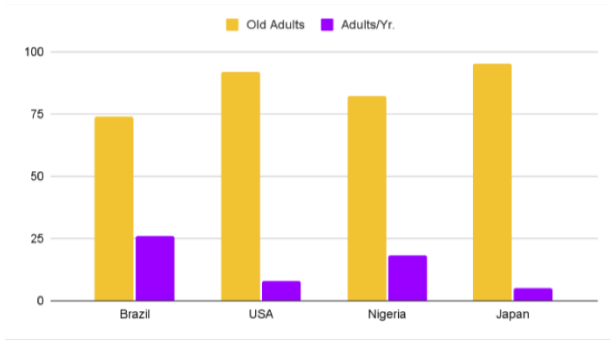


Figure 7: Nationalities Group. Age Analysis Chart

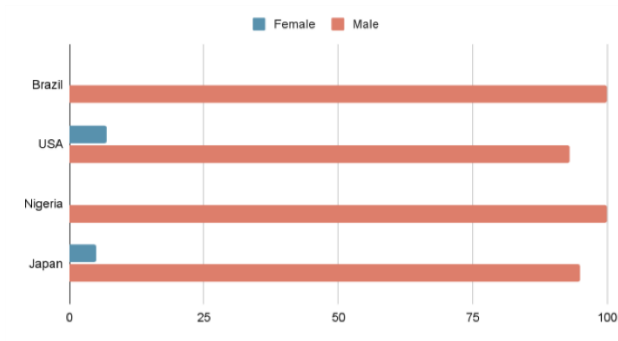


Figure 8: Nationalities Group. Gender Analysis Chart

“old adult male” is portrayed, now with specific cultural hybridities. This becomes evident when we observe only American flags upon adding “USA” to the prompt. Furthermore, there is a complete absence of symbols representing Shintoism (Japan) and Islam (Nigeria), despite these being the most prevalent religions in their respective countries¹⁰. The depicted male figures often come from less affluent backgrounds, particularly in Brazil and Nigeria.

Additionally, there is a tendency to incorporate numerous facial and body tattoos on male figures in Brazil. The consistency in the visual outputs obtained prompts deeper reflections that extend beyond technical issues related to the poor quality of generative AI training data.

The applications of AI Image Generators on a large commercial scale across various sectors such as entertainment, education, and advertising drive developments and research in usability, prompt engineering, and functionalities [22] [37] [38] [39]. However, it is imperative to reflect on how the failure to recognize the inherent social risks and harms of these systems not only limits the ability to conceive alternative futures and/or presents but also reshapes and reinforces social injustices and inequalities [34] [40].

As a critical exercise, the results of this investigation turn the collected images into true visual traces of a power relationship with deep historical roots, precisely in the rise and origins of Western modernity and its colonial expansion worldwide [41]. Thus, analyses of decolonial theories [31] [32] [42] help understand how

¹⁰See footnotes 8 and 9 in Results.



Figure 9: A religious person in (country). Countries: Brazil (top/left); USA (top/right); Nigeria (bottom/left); Japan (bottom/right). 2023. Generated by Midjourney.

data colonialism and contemporary capitalist practices [2] [12] inform how social hierarchies, linguistic ideologies, and generative AI models co-produce [24].

Discourses in HCI and design lack recognition of the ways in which power operates through design. It literally functions as a form of “frozen politics” that reproduces the apparatuses of global and local institutional arrangements, serving as a tool of oppression and rights deprivation [43]. According to Escobar [44], modernity has its roots in Anglo-European traditions with ethno-political foundations in Judeo-Christian theology and Greek/European philosophy. On one hand, these grounds thought in a rationalist tradition that emphasizes individualism, objectivity, science, and economics as separate from human existence. On the other hand, it reinforces ontological dualisms, such as nature/culture, the West/the rest, and subject/object. Therefore, when this becomes the structuring condition of design technology development, it is easier to understand how bias in outputs from the prompt “imagine a religious person” reveals the machine intelligence’s inability to resolve the inherent ontological dualisms in its system. Its visual vocabulary represents the “Western code” of modernity, developed over centuries of colonization of subjectivity and knowledge, bodies, gender, and sexuality, cultural practices, and aesthetics [45].

Addressing these issues requires an “epistemological leap” that moves away from universalizing narratives of knowledge production and contextualizes diverse experiences, epistemologies, and narratives [46]. The decolonization of design practices suggests participatory and inclusive ways of thinking about HCI through de-Westernizing approaches and more humanistic perspectives within practices involving transcultural technology design, humanizing

emerging technology, participatory action research, design anthropology futures, and design tech for social and racial justice [3].

Therefore, situating these initiatives around a plurality of movements and discourses from across the world, as opposed to a one world-system, means positioning the act and thought of design in “in-betweenness” [41] [47]. It involves awakening an understanding of the ability to perceive the contradictions and complexities of the world, distinguishing the inherited from the acquired, learning from the imposed; developing what Anzaldúa calls “facultad” or “conocimiento” [47], and what Tony Fry understands as “border thinking” [41]. Inspired by Escobar’s thinking, Ansari [8] advocates for rethinking design through the lenses of political ecology and an ecological approach that considers coexistence and the experience of the other as the basis for a practice that can potentially enable a plurality of design discourses. Recognizing the importance of these issues creates new territories for design, where insights and paradigm shifts can be tested, identifying, and enabling active and transformative sociotechnical practices that challenge design professionals and theorists.

5 CONCLUSION

This article presents a critical investigation into the default biases produced by AI Image Generators. To conduct this inquiry, the research utilizes the Midjourney platform, employing two distinct prompts: (1) “Imagine a religious person” and (2) “Imagine a religious person in X,” where X represents the countries Brazil, United States, Nigeria, and Japan.

The study delves into a comparative analysis of the biases that arise from these prompt groups. As a critical exercise, the article seeks to unveil how these biases are shaped and perpetuated by the algorithms of Midjourney. Through the insightful application of data colonialism as a conceptual framework, the research sheds light on the power dynamics and asymmetries that underlie the generation of visual content by generative AI models. Thus, it is possible to elucidate the profound impact of social hierarchies and linguistic ideologies on the development of standard visual vocabularies in these models.

The findings of this research go beyond the identification of biases and extend to the exploration of their harmful implications. This understanding is particularly crucial in the contemporary scenario, where AI technologies play an increasingly pervasive role in shaping cultural narratives and perceptions.

It is important to note that this study employs decolonial theories to offer timely reflections for design practice. It underscores the need for a paradigm shift in the design of AI systems, emphasizing the importance of reassessing the processes involved in the development and deployment of these technologies. By considering the implications of bias through a decolonial lens, the research advocates for a more inclusive and culturally sensitive approach to design practice in the context of AI.

In essence, this article not only contributes to the academic discourse on biases in AI Image Generators but also provides actionable insights for practitioners in the fields of HCI and design. The research encourages a critical reexamination of the entire AI image generation ecosystem and emphasizes the importance of expanding discussions, processes, and inclusive practices in mitigating

biases and promoting a more equitable technological landscape for creative practices.

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